

Victor Gaya

AI Research Engineer | Deep Learning & Remote Sensing



Profile

PhD in AI with 3 years of experience developing deep learning models for satellite imagery. Comfortable working on the full ML pipeline, from data preparation to model training and evaluation.

Contact

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🔍 Research Gate: Victor Gaya

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Languages

🇫🇷 French - Native Language

🇬🇧 English - Fluent (TOEIC C1, 2022)

Technical Skills

🔧 Python, PyTorch, TensorFlow, Git

📦 Docker, Linux, Bash

🧠 HuggingFace, scikit-learn, W&B

✍️ LaTeX, MS Office

Involvements

- Drone association founder at CEA
- Concert organiser & technical manager
- Student trips association president
- School annual Gala organiser

Other Interests

🎵 Music production

🎾 Tennis

⚡ Mountaineering

🧗 Climbing

WORK EXPERIENCE

Jan. 2026 -
Today



**Research fellow;
AI & aerospace**

📍 Singapore, Singapore

National University of Singapore, ECE

- Applying **Foundation Models** and **Generative AI** to analyze complex **satellite imagery**.
- Generalizing learned representations across domains via **data-driven approaches**.
- Supervising **interns and students** on experimental design and validating **research outcomes**.

Dec. 2022 -
Dec. 2025



**Junior researcher;
AI & aerospace**

📍 Arpajon, France

Research center

- Developed **Deep Learning** for **SAR interferometry** and **3D reconstruction** in industrial contexts.
- Combined **data fusion** and **auto-supervised learning** based on **statistical modeling**.
- Disseminated results at **international conferences** (oral and poster).

Mar. 2022 -
Sept. 2022



Machine Learning intern

📍 Toulouse, France

French Aerospace Agency

- Implemented a **user feedback system** on a machine learning model for the **detection of anomalies** in military satellite operations.
- Improved the **AI model** deployed on **operational satellite missions**.

May 2021 -
Aug. 2021



Data Science intern

📍 Marseille, France

KORNER (KLANIK)

Consulting in data

- Construction of a **neural network** to detect **Asian hornets and bees** through a **video stream**.
- The **AI model** is now used to build **Asian hornet traps** in France.

EDUCATION

2022 - 2025



PhD Degree

📍 Palaiseau, France

Institut Polytechnique de Paris

Télécom Paris

Worked on deep learning methods to improve SAR satellite images, especially regarding data-fusion (optical and SAR) and interferometry.

2016 - 2022



Engineering - Master Degree

📍 Lille, France

JUNIA

ISEN Lille

Master specialized in AI technology. Music and technology option.